



ARI 410/510: Machine Learning University of Michigan-Flint CIT Fall 2026 Syllabus

Course Overview

Course Description: (from the catalog) Study of machine learning, a subfield of artificial intelligence intersecting with statistics, cognitive science, information theory, optimization, and probability theory. Topics include but are not limited to linear discriminants, neural networks, decision trees, support vector machines, unsupervised learning, and reinforcement learning. With examples from real-world applications, students learn to build systems that learn and adapt from experience.

tl;dr: Math & programming tools that enable computers (***machines***) to ***learn*** from data.

Credits: 4

Instructor:

Prof. Steven Wilson

Email: steverw@umich.edu

Office: MSB 203D

Meeting times:

Tuesday/Thursday

2:30PM - 4:10PM

Location:

MSB 302

Learning objectives:

After completing this course, students should be able to:

- Recall foundational machine learning principles and concepts;
- Explain the mathematical foundations for the field of machine learning;
- Collect and prepare datasets for machine learning applications;
- Use software libraries to train, fine-tune, and evaluate machine learning models;
- Present experimental results to justify model and hyperparameter selection; and
- Build and evaluate end-to-end machine learning systems.

Course format: *Hybrid Synchronous*

Students are registered for both online and in-person sections and are expected to attend class live during the official meeting times. However, you are welcome to switch between modalities on a class-by-class basis without notifying the instructor.

In the case of inclement weather resulting in hazardous road conditions, the instructor may send a message to all students no less than 1 hour before class starts announcing that day's class session will take place online only.

Course communication

Canvas:

- All lecture slides will be posted on Canvas before the start of class.
- All assignment submissions should be made on Canvas unless otherwise noted
- Zoom links for online meetings can be found on Canvas.

Course Discord server: <https://discord.gg/V2UUN2jfg>

- We will use this server for asynchronous communication outside of class.

Instructor Discord username: s_wilson

- Discord is usually the best/fastest way to communicate with me.

Instructor email: steverw@umich.edu

- I aim to reply to emails within 24 hours M-F.
- Please feel free to follow-up if you don't get a reply by then.

Student support hours:

- Drop-in times: Tuesday/Thursday 4:15-5:15pm
- Appointments: Schedule via Discord message
- Location: MSB 203D or [Zoom](#)

Course materials

Textbook:

- For the course, the official textbook will be the [100 page ML book](#) (free ebook)
- You can also find a lot of free ML textbooks [here](#) if you'd like to do some additional reading.
 - One of the most foundational ML books is [this one from Bishop](#)
 - [A \(not free\) book](#) I personally like

Other readings:

- Helpful supplemental readings may be posted on Canvas throughout the semester.

Software:

- Python IDE of your choice (jupyter/colab, vscode, pycharm, emacs, ...)

Assessment

Your grade in the course will be based on a weighted average of items in the following categories:

Category	Weight
Assignments	25%
Problem Sets	15%
Presentations	10%
Project	45%
Participation/engagement	5%

Assignments: there will be around 3-4 lab-type assignments that require you to apply concepts from the course, run experiments, and analyze your results.

In-class Problem sets: These sets of low-stakes problems are designed to help you check and refine your understanding of the *theoretical* and *mathematical* concepts covered in the course. These correspond to the material covered in lectures. They can be completed in person or remotely but some parts must be completed *during the class time* (see below) unless an exception has been prearranged with the instructor. There are several stages to competing these problems:

1. **Solo:** first, you should complete the problems on your own and *submit* your answers before the start of class on the due date.
Graded on completion and effort.
2. **Group:** at the start of class, in small groups, you should come to a consensus about the answers.
Graded based on engagement with group members.
3. **Class:** for each question, one student will randomly be selected to present and justify the solution. The entire class will then ask questions and work together as necessary to come to a solution.
Graded based on participation and effort (not correctness).

4. **Final submission and bonus:** after this, update your original submission with annotations regarding what changed and why. You should list the names of the group members you worked with and certify their engagement in the discussion. Finally, some problem sets have a bonus question that can be completed at this stage. Submit by the start of the following class period.
Graded based on correctness of final answers and thoughtful explanations of changes made.

Project: this will be a group project with a proposal and several checkpoints throughout the semester, and status updates will be presented to the class. The focus of the project is designing and evaluating a realistic ML task, including data, appropriate metrics, and models that are able to do the task.

Presentations: there will be 2 or 3 presentations throughout the semester. Each will have a predefined theme and rubric provided when it is assigned.

Participation/engagement: it is up to you to demonstrate that you are engaged with the course. Most students typically receive all points as long as they consistently engage with the instructor and other students in class either online or offline.

Grading scale

Overall percentage	Letter grade
>93%	A
90% - 92.9%	A-
87% - 89.9%	B+
83% - 86.9%	B
80% - 82.9%	B-
77% - 79.9%	C+
73% - 77%	C
70% - 72.9%	C-
67% - 69.9%	D+
63% - 66.9%	D
60% - 62.9%	D-
< 60%	F

Policies and Expectations

Late work: I try to be flexible about deadlines when possible, though the policy I follow depends on the type of assignment.

- *For lab-style assignments and problem sets:* I want you to be proud of what you submit rather than rushing just to meet the deadline. If you need a little bit more time, don't be afraid to ask. No-penalty extensions will typically be granted as long as you:
(1) notify me *in advance of the due date* (2) propose a new date and time when you can commit to having the work done (3) do not abuse the system (i.e., you are expected to complete *most* assignments on time).
- *For presentations and project milestones:* extensions typically cannot be granted because these require in-class interaction or follow-up on a specific date. However, if something comes up (e.g., you have a conference presentation at the same time as class), please feel free to ask and we can try to work something out that is fair for everyone. This requires that we have a conversation (in person, Discord, email, etc) before the deadline. If I do not hear from you, I will assume that you don't plan to complete the assignment.

Attendance: I hope that you believe coming to class is worth your time and effort, and regular attendance is expected. If you are unable to attend class on a given day, you should still make time to watch the recorded lecture, review any material covered that day, and complete your assignments. You do not need instructor permission. It is your responsibility to demonstrate that you are engaging with the course content. However, students who *consistently* fail to show up to the course (either in person or online) without prior permission/communication with the instructor may have their final grade reduced by up to 1 full letter grade.

Generative AI: This is a course about AI. You may use AI tools like ChatGPT or Claude to help you in the course, just like you would be able to in the “real world”. All I ask is that you *disclose when and how you use these tools*. This can be in the form of a note on your assignment reports, presentations, or project updates, or even by sharing the chat history from your chat session. You take full responsibility for anything you submit regardless of what tools were used. AI detection tools are biased, unreliable and will never be used to “catch” students. However, the instructor reserves the right to deduct points for extremely bland, overly verbose, or painfully generic writing and presentations, whether they are written by Generative AI tools or not. Finally, and most importantly: **do not let AI tools become a substitute for thinking, learning, or interacting with your fellow humans.**

Sample Schedule

Note that this is just an example to give you an idea about how the semester might play out, and it is subject to change. The official schedule will be released throughout the semester on Canvas.

Week 1 : Course Intro

- About the course
- Intro to ML

Week 2 : ML Basics 1

- Notation and definitions
- Fundamental algorithms

Week 3 : ML Basics 2

- Regression
- Regression (cont.)

Week 4 : Data

- Annotation
- Annotation (cont.)

Week 5 : Unsupervised Learning 1

- Intro to Unsupervised Learning
- Intro to Unsupervised Learning (cont.)

Week 6 : Supervised Learning 1

- Feature Selection
- Decision Trees

Week 7 : Supervised Learning 2

- Random Forests
- Multimodal Data

Week 8: Supervised Learning 3

- Supervised Learning: Evaluation

- Project Phase 1 Presentations

Week 9: Neural Networks 1

- No class: Fall Break
- Neural Networks

Week 10 : Neural Networks 2

- Neural Networks (cont.)
- Transformers

Week 11 : Neural Networks 3

- Transformers (cont.)
- ML Practice: Experimental Design

Week 12 : Reinforcement Learning

- Intro to RL
- Deep RL

Week 13 : LLMs

- Project Phase 2 Presentations
- Supervised LM + RL = LLM

Week 14 : Contrastive Learning

- Contrastive Learning: CLIP
- No class: Thanksgiving

Week 15 : MLOps

- MLOps: Intro to MLOps
- ML Research Topics

All the rest...

Below (highlighted in grey) is some standard language that you have probably seen in many other UM-Flint syllabi. It is important, and still pertains to this course, but I moved it to the end since you have likely read it before. If you haven't, please do take a moment to check it out now.

1. Syllabus Statement

This course syllabus is a guide and every attempt is made to provide an accurate overview of the course. Circumstances and events may make it necessary for the instructor to modify the syllabus over the course of the semester. If you have any questions or concerns about revisions or changes made to the syllabus, please consult with your instructor.

2. Land Acknowledgement

We would like to acknowledge that the land the University of Michigan-Flint occupies is the ancestral, traditional, and contemporary homeland of many Indigenous nations, most recently the Anishinabek (including Potawatomi, Chippewa/Ojibwe, and Odawa) tribal nations.

We acknowledge the painful history of genocide, forced relocation, and removal of many from this territory, and we honor and respect the many Indigenous people, including those of the Three Fires Alliance, who are still connected to the land upon which the University of Michigan-Flint is located.

For more information: <https://www.umflint.edu/icc/>

3. UM-Flint's Statement on Mutual Respect

As part of the University of Michigan, our University strives to provide an equitable, safe space for all individuals. This class will be conducted in an atmosphere of mutual respect. I encourage your active participation and welcome both respectful discourse and reasoned debate. However, if your language or conduct at any time demonstrates a lack of respect for anyone's race, gender identity or expression, sexuality, culture, beliefs, or abilities, you will not be permitted to participate further. -Adapted from "LGBTQ On Campus for Faculty & Staff"

For more information: <https://www.umflint.edu/cgs/> and <https://www.umflint.edu/dei/>

4. *Academic Integrity

The University of Michigan-Flint has established a Code of Academic Conduct as well as a statement regarding the Procedural Rights of the Accused Student. You may visit <http://tinyurl.com/umf-academic-integrity>

5. *Disability Statement:

The University of Michigan-Flint is committed to fostering an inclusive and accessible learning environment for all students. We recognize that students with disabilities may require specific accommodations to fully participate in our courses and reach their academic potential. If you have a disability or suspect

that you may have one, we encourage you to seek the necessary support and accommodations from the Disability & Accessibility Support Services Office (DASS).

DASS is dedicated to working collaboratively with students to identify and implement appropriate accommodations based on individual needs. Some examples of accommodations may include extended time for exams, note-taking assistance, alternative formats for course materials, or classroom modifications. These accommodations are designed to level the playing field and ensure equal opportunities for success. DASS is located in suite 264 of the University Center and can be contacted at (810) 762-3456 or dassflint@umich.edu.

Accommodation requests can be initiated online at <https://www.umflint.edu/disabilitysupportservices/>.

6. Resources for Student Well-Being

As students, you may experience stressors that can impact both your academic experience and your personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For both academic and personal concerns, UM-Flint offers the following [resources](#).